THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 5: **Arithmetic Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The 12th term of the AP whose 1st term is 7 and 5th term is 19 is: 1

(a) 36 **(b)** 43 **(c)** 40 **(d)** 37

2. The number of terms in the AP $5, 9, \ldots, 97$ is: 1

(a) 24 **(b)** 26 **(c)** 25 **(d)** 23

3. The sum of all two-digit multiples of 3 is: 1

(a) 1566 **(b)** 1653 **(c)** 1665 **(d)** 1683

4. The sum of the first 14 terms of the AP $12, 17, 22, 27, \ldots$ is: 1

(a) 705 **(b)** 546 **(c)** 623 **(d)** 1246

5. The middle term of an AP with 13 terms, first term 5 and last term 41, is: 1

(a) 26 **(b)** 23 **(c)** 20 **(d)** 41

6. If $a_8 = 25$ and $d = 3$, then $a =$ 1

(a) 4 **(b)** 1 **(c)** 46 **(d)** 7

7. The 9th term of the AP whose 1st term is 4 and 4th term is 22 is: 1

(a) 52 **(b)** 58 **(c)** 46 **(d)** 54

8. The number of terms in the AP $8, 11, \ldots, 98$ is: 1

(a) 30 **(b)** 32 **(c)** 31 **(d)** 33

9. The sum of all two-digit multiples of 4 is: 1

(a) 1200 **(b)** 1092 **(c)** 1188 **(d)** 1288

10. The sum of the first 20 terms of the AP $-6, -2, 2, 6, \ldots$ is: 1

(a) 640 **(b)** 570 **(c)** 714 **(d)** 1280

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Which term of the AP $7, 13, 19, 25, \ldots$ is 127? 2

12. Find the sum of the first 18 terms of the AP $14, 20, 26, 32, \ldots$ 2

13. Find the sum of all natural numbers between 1 and 100 that are divisible by 4. 2

14. Which term of the AP $11, 15, 19, 23, \ldots$ is 79? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In an AP, the 3rd term is 19 and the 8th term is 54. Find the AP and its 15th term. **3**

16. Find the sum of the first 20 terms of the AP whose nth term is $a_n = 3n + 2$. **3**

17. The sum of the first 10 terms of an AP is 185 and the first term is 5. Find d and the 10th term. **3**

OR

In an AP, the 2nd term is 2 and the 7th term is 32. Find the AP and its 12th term. **3**

18. Find the sum of the first 15 terms of the AP whose nth term is $a_n = 5n - 1$. **3**

OR

The sum of the first 12 terms of an AP is 360 and the first term is 8. Find d and the 12th term. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. How many terms of the AP $54, 50, 46, 42, \ldots$ must be taken so that the sum is 360? **5**

OR

A ladder has 11 rungs whose lengths decrease uniformly from 45 cm at the bottom to 25 cm at the top. Find the length of the wood needed for the rungs and the length of the middle rung. **5**

20. How many terms of the AP $12, 18, 24, 30, \ldots$ must be taken so that the sum is 714? **5**

OR

A ladder has 13 rungs whose lengths decrease uniformly from 60 cm at the bottom to 30 cm at the top. Find the length of the wood needed for the rungs and the length of the middle rung. **5**

END OF QUESTION PAPER — Total: 40 Marks

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